

CSE 4631

Digital Signal Processing

Week 6

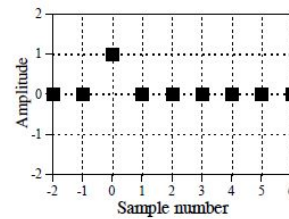
Properties of Convolution

Common Impulse Responses: **Delta Function**

- An impulse on the input produces an identical impulse on the output.
- *All signals pass through the system without any change.*
- Convolving any signal with a delta function results in exactly the same signal. Mathematically, this is written as: $x[n] \square \delta[n] = x[n]$
- This property makes the delta function the **identity** for convolution.
- If the delta function is made larger or smaller in amplitude, the resulting system is an **amplifier** or **attenuator**, respectively. Mathematically: $x[n] \square k\delta[n] = kx[n]$
- If the impulse response is a delta function with a shift, then the system introduces an identical shift between the input signal and the output signal. This can be described as a signal **delay** or a signal **advance**. Mathematically: $x[n] \square \delta[n+s] = x[n+s]$

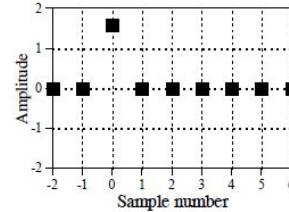
a. Identity

The delta function is the *identity* for convolution. Convolving a signal with the delta function leaves the signal unchanged. This is the goal of systems that transmit or store signals.



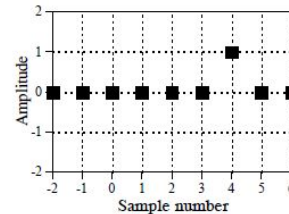
b. Amplification & Attenuation

Increasing or decreasing the amplitude of the delta function forms an impulse response that *amplifies* or *attenuates*, respectively. This impulse response will amplify the signal by 1.6.



c. Shift

Shifting the delta function produces a corresponding shift between the input and output signals. Depending on the direction, this can be called a *delay* or an *advance*. This impulse response delays the signal by four samples.



d. Echo

A delta function plus a shifted and scaled delta function results in an *echo* being added to the original signal. In this example, the echo is delayed by four samples and has an amplitude of 60% of the original signal.

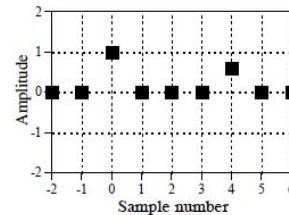


Figure: Simple impulse responses using shifted and scaled delta functions

Properties of Convolution (contd.)

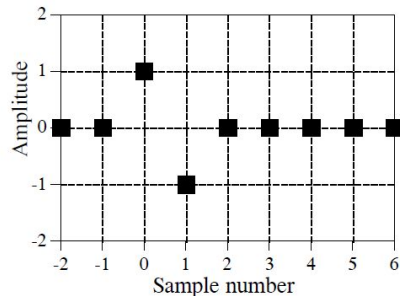
Calculus-like Operations

- Convolution can change signals in ways that resemble integration and differentiation.
- The discrete operation that mimics the *first derivative* is called the **first difference**.
- The discrete form of *integral* is called the **running sum**.

Figure: Impulse responses that mimic calculus operations

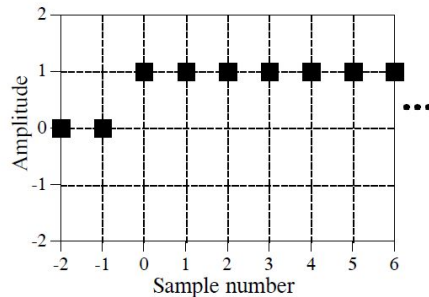
a. First Difference

This is the discrete version of the *first derivative*. Each sample in the output signal is equal to the *difference* between adjacent samples in the input signal. In other words, the output signal is the *slope* of the input signal.



b. Running Sum

The running sum is the discrete version of the *integral*. Each sample in the output signal is equal to the sum of all samples in the input signal to the *left*. Note that the impulse response extends to infinity, a rather nasty feature.



$$x(n) = \{x[0], x[1], x[2], x[3], \dots\}$$

$$h_{fd}(n) = \{\dots, 0, \mathbf{1}, -1, 0, 0, \dots\}$$

$$h_{rs}(n) = \{\dots, 0, \mathbf{1}, 1, 1, \dots\}$$

$x(n)$			$x[0]$	$x[1]$	$x[2]$	$x[3]$
$y[0] = x[0]$	0	-1	1	0	0	0
$y[1] = x[1] - x[0]$	0	0	-1	1	0	0
$y[2] = x[2] - x[1]$	0	0	0	-1	1	0
$y[3] = x[3] - x[2]$	0	0	0	0	-1	1

Table: Convolution table for **first difference**

$x(n)$			$x[0]$	$x[1]$	$x[2]$	$x[3]$
$y[0] = x[0]$	1	1	1	0	0	0
$y[1] = x[1] + x[0] = x[1] + y[0]$	1	1	1	1	0	0
$y[2] = x[2] + x[1] + x[0] = x[2] + y[1]$	1	1	1	1	1	0
$y[3] = x[3] + x[2] + x[1] + x[0] = x[3] + y[2]$	1	1	1	1	1	1

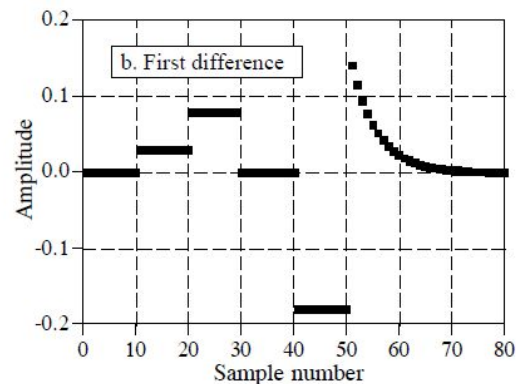
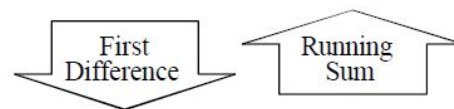
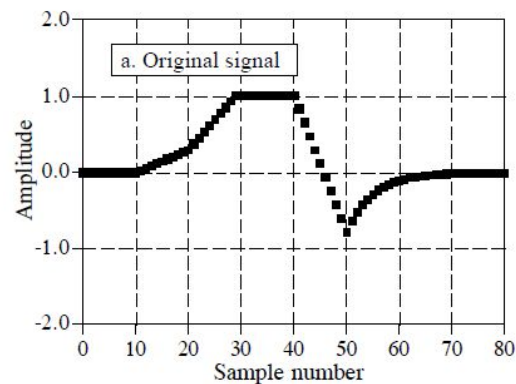
Table: Convolution table for **running sum**

Properties of Convolution (contd.)

Calculus-like Operations (contd.)

- Calculation of the first difference can be done using: $y[n] = x[n] - x[n-1]$
- Calculation of the running sum can be done using: $y[n] = x[n] + y[n-1]$
- These relations are called recursion equations or difference equations

FIGURE 7-3
 Example of calculus-like operations. The signal in (b) is the *first difference* of the signal in (a). Correspondingly, the signal in (a) is the *running sum* of the signal in (b). These processing methods are used with discrete signals the same as differentiation and integration are used with continuous signals.



Properties of Convolution (contd.)

Low-pass Filters

- Filter kernels (another name for filter's impulse response) are composed of a group of *adjacent positive points*.
- Results in each sample in the output signal being a **weighted average of many adjacent points from the input signal**. This average *smoothes* the signal, thereby removing high-frequency components.
- While delta function impulse response passes the entire signal, it *passes only the low-frequency components*.
- Usages: noise reduction, signal separation, wave shaping, etc.

Properties of Convolution (contd.)

Low-pass Filters (contd.)

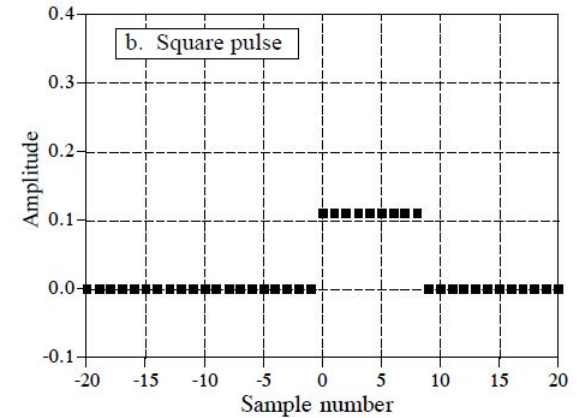
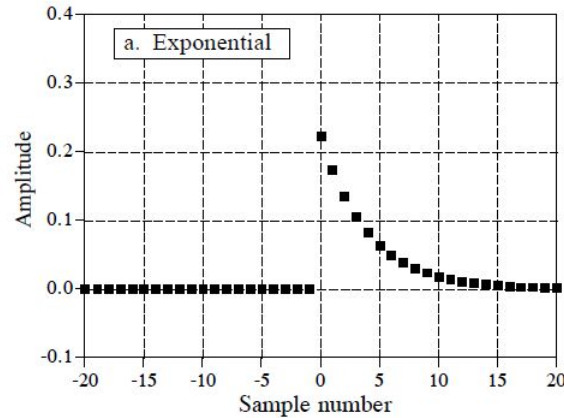
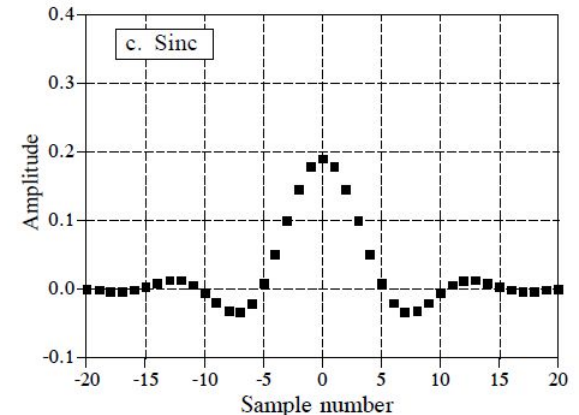


FIGURE 7-4

Typical low-pass filter kernels. Low-pass filter kernels are formed from a group of adjacent positive points that provide an averaging (smoothing) of the signal. As discussed in later chapters, each of these filter kernels is best for a particular purpose. The exponential, (a), is the simplest recursive filter. The rectangular pulse, (b), is best at reducing noise while maintaining edge sharpness. The sinc function in (c), a curve of the form: $\sin(x)/x$, is used to separate one band of frequencies from another.



Properties of Convolution (contd.)

High-pass Filters

- Filter kernels are composed of *a spike surrounded by many adjacent negative points*.
- Formed by subtracting the corresponding low-pass filter kernels from a delta function.
- Since the delta function impulse response passes the entire signal, and low-pass filters pass only the low-frequency components, by superposition, a filter kernel consisting of a delta function minus the low-pass filter kernel will pass the entire signal minus the low-frequency components.
- It is a common strategy to devise the low-pass filter first and then transform it to a high-pass filter.

Properties of Convolution (contd.)

High-pass Filters (contd.)

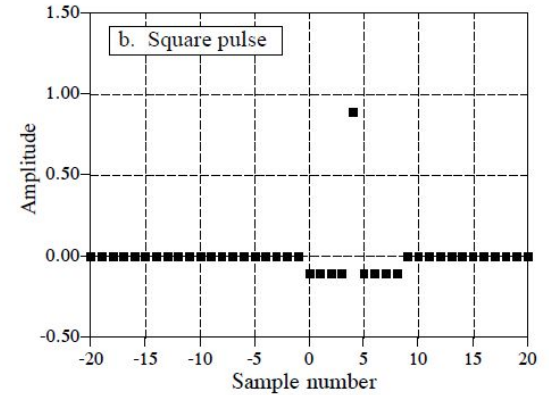
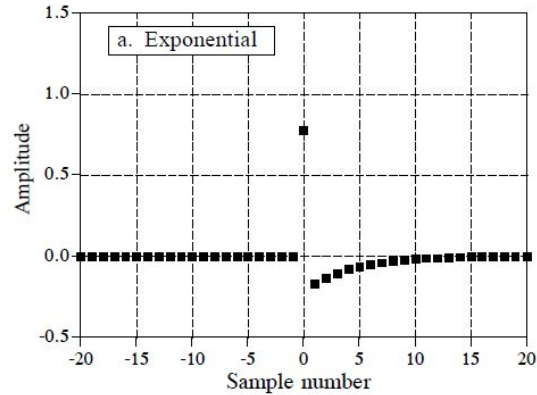
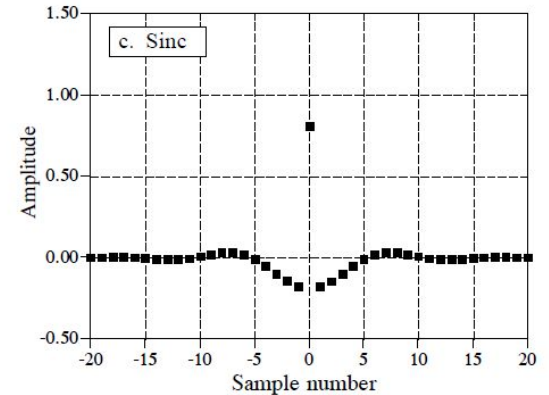


FIGURE 7-5

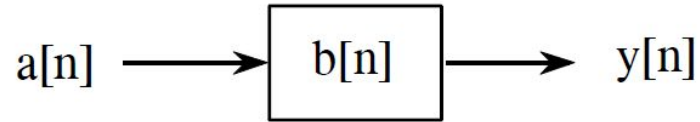
Typical high-pass filter kernels. These are formed by subtracting the corresponding low-pass filter kernels in Fig. 7-4 from a delta function. The distinguishing characteristic of high-pass filter kernels is a spike surrounded by many adjacent negative samples.



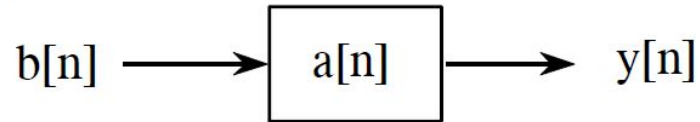
Mathematical Properties of Convolution

Commutative Property: $a[n] \square b[n] = b[n] \square a[n]$

IF



THEN

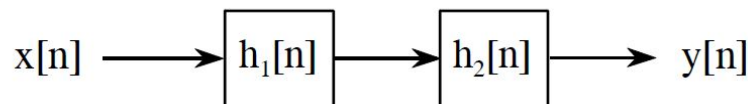


Mathematical Properties of Convolution (contd.)

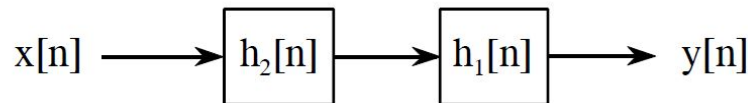
Associative Property: $(a[n] \square b[n]) \square c[n] = a[n] \square (b[n] \square c[n])$

- Provides two important characteristics of cascaded linear systems:
 - The order of the systems can be rearranged without changing the overall response of the cascade
 - Any number of cascaded systems can be replaced with a single system. The impulse response of the replacement system is found by convolving the impulse responses of all of the original systems.

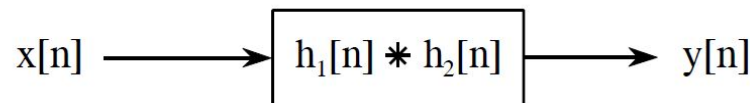
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ALSO

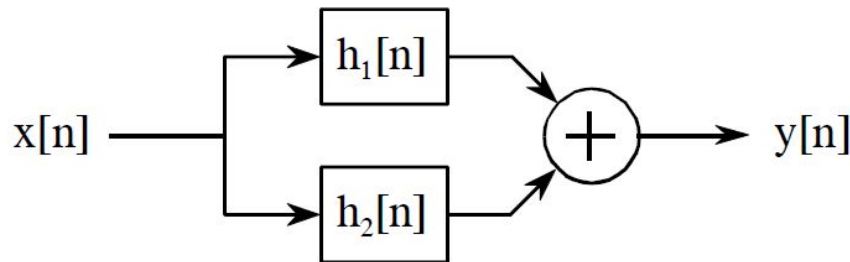


Mathematical Properties of Convolution (contd.)

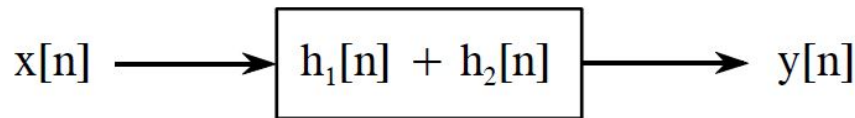
Distributive Property: $a[n] \square b[n] + a[n] \square c[n] = a[n] \square (b[n] + c[n])$

- Describes the operation of parallel systems with added outputs.
 - Any number of parallel systems can be replaced with a single system. The impulse response of the replacement system is equal to the sum of the impulse responses of all of the original systems.

IF



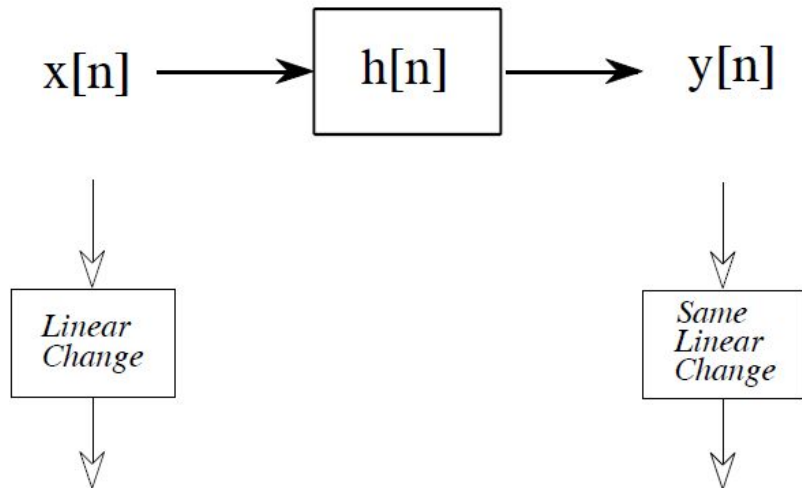
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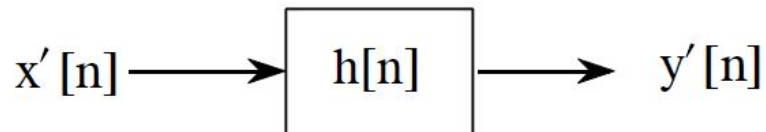
Mathematical Properties of Convolution (contd.)

Transference between the Input and Output

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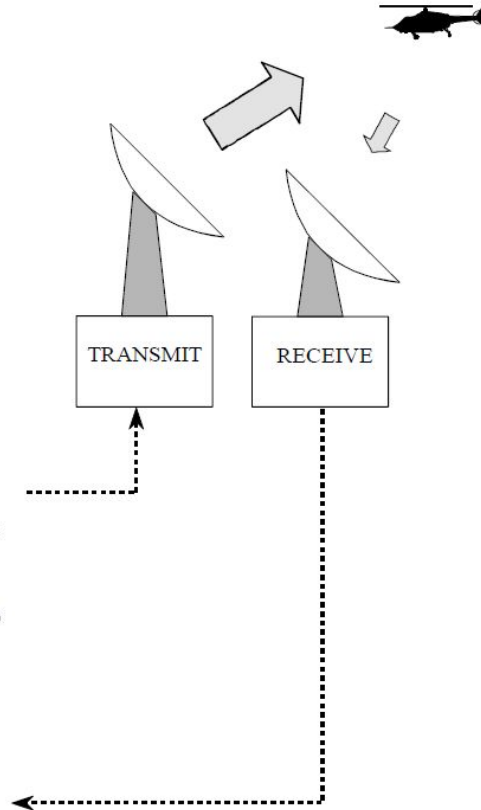
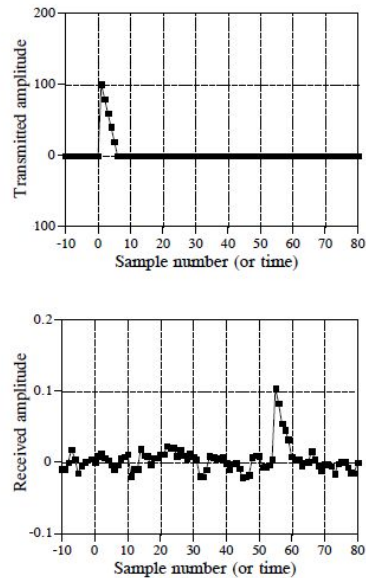


Correlation

- Determines the occurrence of a signal in another signal. Used to detect a known waveform in a noisy background.
- Very similar to convolution: Uses two signals to produce a third signal
- This third signal is called the **cross-correlation** of the two input signals (target signal and received signal)
- If a signal is correlation with itself, the resulting signal is instead called the **autocorrelation**
- The amplitude of each sample in the cross-correlation signal is a measure of how much the received signal resembles the target signal, at that location.
- A peak will occur in the cross-correlation signal for every target signal that is present in the received signal.
- The peak is symmetrical between its left and right even if the target signal isn't symmetrical
- The width of the peak is twice the width of the target signal

FIGURE 7-13

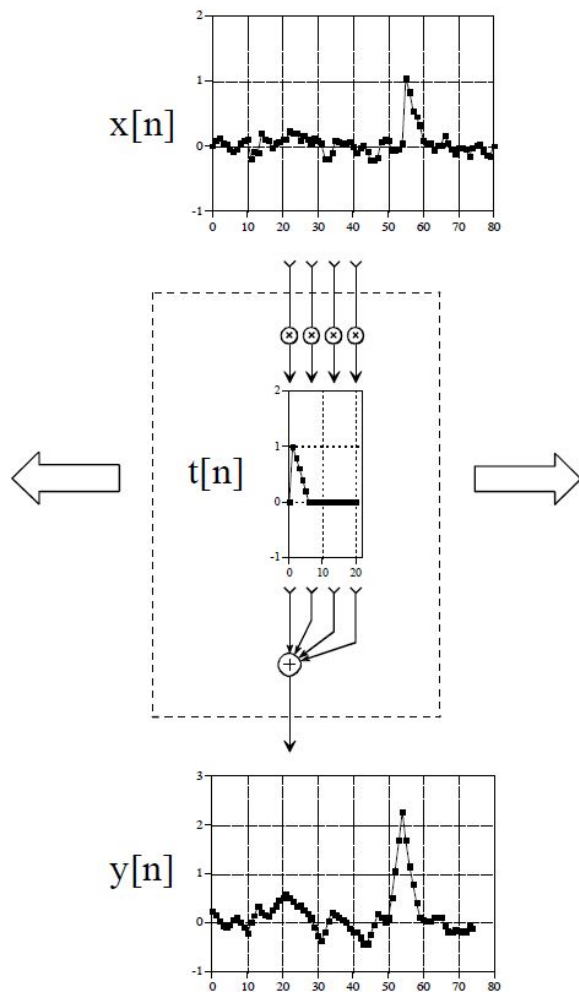
Key elements of a radar system. Like other echo location systems, radar transmits a short pulse of energy that is reflected by objects being examined. This makes the received waveform a shifted version of the transmitted waveform, plus random noise. Detection of a known waveform in a noisy signal is the fundamental problem in echo location. The answer to this problem is *correlation*.



Correlation (contd.)

- Using correlation to detect a known waveform is frequently called **matched filtering**.
- The correlation machine and convolution machine are identical, except for *one* small difference. Unlike convolution machine, the **signal inside the correlation machine isn't folded** (flipped left-to-right), and the **samples run in the normal direction**.
- Therefore, it can be mathematically written as: $a[n] \square b[-n] = c[n]$

Figure: The correlation machine



Speed of Convolution

- Executing a convolution program includes large number of multiplications and additions required by the algorithm, resulting in long execution time.
- This multiply-accumulate operation is a basic building block of DSP. The speed of DSP computers is often specified by how long it takes to perform a multiply-accumulate operation.
- If a signal composed of N samples is convolved with a signal composed of M samples, ~~$N \times M$ multiply-accumulations~~ must be performed.

Speed of Convolution (contd.)

Three ways of handling the problem of excessive execution time:

- **Keeping the signals as short as possible and using integers instead of floating point**
 - Suitable if you need to run the convolution a few times only
 - Has the best trade-off between execution time and programming effort.
- **Using a computer designed for DSP**
 - Suitable if you need to run the convolution many times, such as in the design of commercial products
 - Example: DSP microprocessors with multiply-accumulate times of only a few tens of nanoseconds
- **Using a better algorithm, such as *FFT convolution***
 - For signals with thousands of samples, FFT convolution can be hundreds of times faster
 - The disadvantage of this algorithm is program complexity